

TrES-2b

R-1196

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190501 · created 2026-07-31T06:06:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458604.86038 +/- 0.0005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1464 +/- 0.0076
Transit depth (Rp/Rs)^2	2.14 +/- 0.22 [%]
Orbital Inclination (inc)	84.29 +/- 0.34
Airmass coefficient 1 (a1)	1.211 +/- 0.011
Airmass coefficient 2 (a2)	-0.1591 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	None
Optimal Aperture	3.77
Optimal Annulus	6.92
Transit Duration (day)	0.0834 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1464 ± 0.0076 vs 0.1254 (deviation 1.8σ)
Duration fit vs published	0.0834 d vs 0.0746 d (deviation 1.3σ)
Mid-time O–C	1.6 min
Depth SNR	12.5
Residual scatter	0.92 %

Light curve

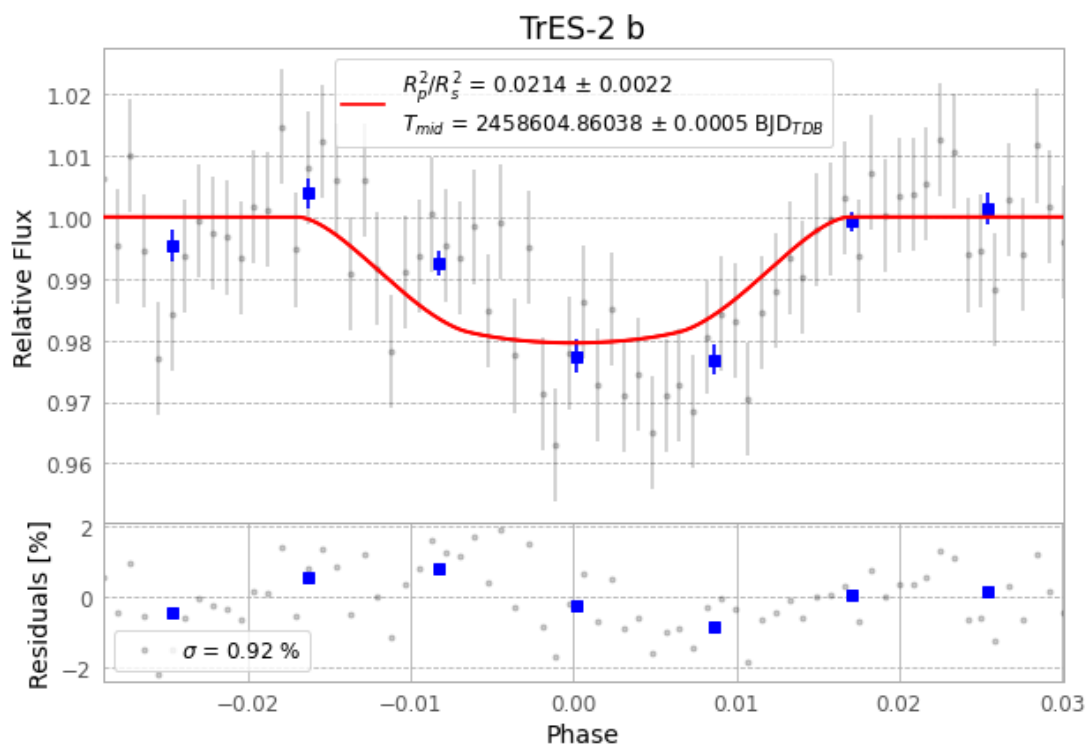


Figure 1 — FinalLightCurve_TrES-2 b_2019-04-30.png (SHA-256 e69990ba8e45a38c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 96], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 3ef556c57bf4c93c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-2190501065412.FITS → TRES-2190501102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-TRES-2-190501.log (5347ff575ff1...), FinalLightCurve_TrES-2 b_2019-04-30.png (e69990ba8e45...), FinalLightCurve_TrES-2 b_2019-04-30.csv (01a33886c657...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:06Z.